Supplementary Material for HKVM-RAG: Key-Value-Separated Hypergraph Evidence Organization for Multi-Hop RAG

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I. Scope of the Supplementary Material

This supplementary material contains reproducibility and diagnostic evidence that supports, but is not necessary to state, the main paper's claims. It follows the paper's bounded framing: HKVM-RAG is evaluated as a fixed-substrate evidence-indexing and dense-aware evidence-control method, not as a hidden-test benchmark, a standalone dense-retrieval replacement, or a full reproduction of every related RAG system. All tables below are derived from frozen prediction files and paper-facing exports. No new LLM extraction calls are introduced by these supplementary analyses.

II. Source-Level Significance

Table II gives the paired-bootstrap support for the first-stage source diagnostics summarized in the main paper. The confidence interval tests WHG-KV against the corresponding first-stage baseline. The final column reports the point-estimate margin between WHG-KV and the strongest non-WHG structured complement in the same dataset–first-stage block.

The source-level exports use seeds 13, 29, and 43, 5,000 paper-export bootstrap samples per dataset–first-stage row, frozen train/dev prediction files, and no new LLM extraction calls. The artifact README records the ColBERTv2, BM25, Contriever, and HotpotQA source-level export roots, hashes, and redistributability notes.

III. Dense-Aware Controller Significance

The main paper reports paired-bootstrap F1 differences for the dense-aware HKVM controller against ColBERTv2 and learned HKVM calibration. Table III complements the main paper by reporting the corresponding EM differences with 95% confidence intervals. The controller improves EM over both ColBERTv2 and learned HKVM calibration on all three benchmarks. The 2Wiki gain over learned HKVM is intentionally small but positive, because learned HKVM calibration and the HKVM-only controller are already strong on that dataset.

The F1 counterparts of these intervals appear in the main paper's controller bootstrap table. Controller bootstrap exports use 5,000 samples over frozen prediction files with out-of-fold HKVM training signals. The source-level HotpotQA ColBERTv2 row in Table II is a separate source-replacement export, which explains its slightly different confidence interval from the main controller-vs-ColBERTv2 interval.

IV. Evidence-Selection Trajectory

Table IV records the trajectory from no-gold WHG retrieval to oracle answer-path selection, train-derived support scoring, learned HKVM calibration, and the dense-aware controller. This table is a diagnostic trajectory rather than a single deployed pipeline. The oracle rows use support information and serve only to measure headroom.

The trajectory combines fixed-substrate dev outputs, oracle diagnostics, train-to-dev support-scoring outputs, learned HKVM calibration, and dense-aware controller exports. It should be read as a mechanism-oriented evidence path rather than as a single chronological production pipeline.

V. Standalone Boundary on HotpotQA

HotpotQA is included to prevent a standalone-retrieval overclaim. In the fixed-substrate standalone matrix, WHG-KV improves support coverage over KG-PPR but does not improve answer F1. Table V separates that standalone boundary from the controller-mediated recovery reported in the main paper.

This boundary is compatible with the dense-aware controller result: raw structured retrieval can underperform lexical and dense baselines on HotpotQA, while WHG-KV rank/score signals can still help a learned controller reorder answer-bearing passages.

VI. Negative and Boundary Diagnostics

A. Naive Fusion

MuSiQue contains genuine ColBERTv2–WHG complementarity, but simple fixed-budget fusion and shallow single-feature gates did not reliably exceed ColBERTv2. RRF fusion of ColBERTv2 and HKVM predictions improved F1 over ColBERTv2 by only +0.110 on MuSiQue, far below the learned controller's +6.763. On 2WikiMultiHopQA and HotpotQA,

TABLE I

SUPPLEMENTARY EVIDENCE MAP. THE SUPPLEMENT RECORDS STATISTICAL SUPPORT, BOUNDARY DIAGNOSTICS, AND ARTIFACT PROVENANCE THAT WOULD OVERBURDEN THE MAIN PAPER; IT DOES NOT INTRODUCE ADDITIONAL BENCHMARK CLAIMS BEYOND THE FROZEN RESULT MATRIX.

Supplement section	Supports	Does not support
Source-level significance	WHG-KV-specific complementarity under the tested first-stage sources	Generic benefit from arbitrary structured sources
Controller significance	Dense-aware controller gains over ColBERTv2 and learned HKVM calibration	Hidden-test or deployment-throughput claims
Evidence-selection trajectory	Support-selection bottleneck and learned recovery path	Treating oracle rows as deployable systems
HotpotQA boundary	Separation of standalone failure and controller recovery	Standalone WHG-KV superiority on HotpotQA
Negative diagnostics	Need for learned control and cache-manifest discipline	Claim that LLM triples dominate all OpenIE alternatives
Artifact provenance	Audit path for paper-facing result families	Redistribution of non-redistributable provider or dataset artifacts

TABLE II

SOURCE-LEVEL BASELINE-VS-WHG PAIRED-BOOTSTRAP SUPPORT AND NON-WHG MARGINS. THE CONFIDENCE INTERVAL TESTS WHG-KV AGAINST THE CORRESPONDING FIRST-STAGE BASELINE; THE FINAL COLUMN IS A POINT-ESTIMATE MARGIN OVER THE STRONGEST MATCHED NON-WHG SOURCE. THE TABLE DOES NOT IMPLY THAT ALL STRUCTURED SOURCES HELP OR THAT THE DIAGNOSTICS COVER ALL RETRIEVER FAMILIES.

Dataset	First-stage	WHG-KV ΔF1 vs base	95% CI	Margin over best non-WHG
2WikiMultiHopQA	ColBERTv2	$+11.084$	$[+10.722, +11.442]$	$+8.077$
2WikiMultiHopQA	BM25	$+16.452$	$[+16.039, +16.865]$	$+9.786$
2WikiMultiHopQA	Contriever	$+23.119$	$[+22.668, +23.577]$	$+8.845$
MuSiQue	ColBERTv2	$+6.763$	$[+5.813, +7.753]$	$+8.219$
MuSiQue	BM25	$+12.038$	$[+10.935, +13.120]$	$+12.212$
MuSiQue	Contriever	$+7.813$	$[+6.810, +8.814]$	$+7.605$
HotpotQA	ColBERTv2	$+5.966$	$[+5.479, +6.456]$	$+5.397$
HotpotQA	BM25	$+4.971$	$[+4.554, +5.374]$	$+4.188$
HotpotQA	Contriever	$+5.166$	$[+4.683, +5.634]$	$+5.522$

TABLE III

DENSE-AWARE CONTROLLER EM SIGNIFICANCE. THIS TABLE SUPPLEMENTS THE F1 BOOTSTRAP TABLE IN THE MAIN PAPER WITH EXACT-MATCH (EM) PAIRED-BOOTSTRAP RESULTS.

Dataset	Comparator	ΔEM	95% CI
2Wiki	ColBERTv2	$+11.463$	$[+11.116, +11.820]$
2Wiki	Learned HKVM	$+0.477$	$[+0.388, +0.565]$
MuSiQue	ColBERTv2	$+6.964$	$[+5.984, +7.937]$
MuSiQue	Learned HKVM	$+8.509$	$[+7.509, +9.492]$
HotpotQA	ColBERTv2	$+5.991$	$[+5.510, +6.482]$
HotpotQA	Learned HKVM	$+2.921$	$[+2.579, +3.255]$

RRF improved by +8.800 and +3.379 F1 respectively, but remained below the learned controller (+11.084 and +5.966). Three additional fixed-budget quota mixtures (dense3+hkvm2, dense4+hkvm1) were evaluated on MuSiQue and produced F1 values of 58.237, 58.309, and 57.923, none reliably exceeding ColBERTv2 (58.309). This motivates the learned passage-level controller used in the main paper, and blocks the simpler claim that generic RRF or quota fusion is enough.

B. Controller Feature Ablation

The dense-aware controller uses per-passage feature vectors containing ColBERTv2 and HKVM rank/score signals. Table VI decomposes the full feature set to identify which signal groups carry the controller gain. The ablation keeps the controller protocol, train-to-dev split, and hyperparameters fixed while varying the feature subset supplied to the logistic ranker.

The main finding is that explicit dense-HKVM interaction features are not necessary for the MuSiQue gain. The no-interaction variant reaches 65.738 F1 on MuSiQue, marginally exceeding the full controller (65.073). Score-only (65.254) also approaches full performance. Interaction-only (60.295) is far below full, confirming that interaction features alone are insufficient. On 2Wiki, HKVM calibration is already strong (88.391), so all dense+HKVM variants cluster within 0.7 F1 of each other. The safe mechanism claim is that dense and WHG-KV rank/score signals jointly carry useful passage-reordering information; explicit interaction features help on 2Wiki but are not the primary driver of the MuSiQue gain. This ablation uses out-of-fold HKVM training signals, seeds 13/29/43, 5,000-sample paired bootstrap, and no new LLM extraction calls.

C. Extraction Substrate

The shared LLM-triple cache is a controlled extraction substrate, not a claim that LLM triples dominate all OpenIE alternatives. In extraction-quality diagnostics, LLM triples were much sparser than heuristic OpenIE: objects per passage changed from 34.814 to 4.249 on 2Wiki and from 47.056 to 4.878 on MuSiQue. Pairwise KG-PPR with the sparse LLM tuple cache did not dominate heuristic OpenIE KG-PPR in that diagnostic. The paper therefore treats cache changes as new manifests rather than mixing them into the frozen result matrix.

D. Hyperparameter Sensitivity

Table VII reports WHG-KV sensitivity to the three main hyperparameter groups: the confidence-weight coefficients $\lambda = (\lambda_f, \lambda_s, \lambda_b)$, the weight-scale parameter β, and the diffusion parameter ρ. Each variant changes only one parameter from the default $(\lambda_f{=}0.20, \lambda_s{=}0.40, \lambda_b{=}0.40,\ \beta{=}2.0,\ \rho{=}0.35)$, reuses the existing DeepSeek evidence tuple cache, and evaluates on the MuSiQue dev answerable subset with seed 13.

TABLE IV

EVIDENCE-SELECTION TRAJECTORY FROM NO-GOLD WHG RETRIEVAL TO DENSE-AWARE CONTROL. THE TABLE SUPPORTS THE INTERPRETATION THAT SUPPORT SELECTION IS A REPAIRABLE BOTTLENECK; ORACLE ROWS ARE DIAGNOSTIC UPPER BOUNDS AND NOT DEPLOYABLE MODEL RESULTS.

	2WikiMultiHopQA			MuSiQue			HotpotQA		
Variant	F1	EM	AR@10	F1	EM	AR@10	F1	EM	AR@10
No-gold WHG retrieval	79.299	78.650	99.650	39.925	39.222	38.643	72.957	72.586	98.190
Oracle answer-path selection	90.884	90.832	99.634	74.330	74.059	73.066	91.111	91.047	98.190
Train-derived support scorer	87.212	86.967	98.378	54.252	53.662	50.228	81.525	81.269	96.016
Learned HKVM calibration	88.391	88.181	99.030	56.754	56.213	53.303	82.929	82.705	96.439
Dense-aware HKVM controller	88.846	88.658	100.000	65.073	64.722	64.115	85.810	85.627	100.000

TABLE V

HOTPOTQA STANDALONE BOUNDARY UNDER MATCHED CANDIDATE-PASSAGE BUDGETS. THE TABLE BLOCKS A STANDALONE WHG-KV SUPERIORITY CLAIM WHILE REMAINING COMPATIBLE WITH CONTROLLER-MEDIATED RECOVERY.

Method	AR@10	F1	EM
BM25	99.851	81.835	81.648
Contriever	100.000	79.930	79.730
ColBERTv2	100.000	79.844	79.635
KG-PPR	95.381	73.645	73.315
Static-HG	98.190	73.340	73.018
Weighted-HG-KV	98.190	72.957	72.586

TABLE VI

DENSE-AWARE CONTROLLER FEATURE ABLATION. FULL USES BOTH SOURCE SIGNALS PLUS EXPLICIT DENSE-HKVM INTERACTION FEATURES; NO-INTERACTION DROPS THE EXPLICIT INTERACTION TERMS; SCORE-ONLY AND RANK-ONLY KEEP ONLY THOSE FEATURE GROUPS; INTERACTION-ONLY USES ONLY THE EXPLICIT CROSS-SOURCE FEATURES. COLBERTV2 AND LEARNED HKVM CALIBRATION ROWS ARE REFERENCE PREDICTORS INCLUDED IN THE MAIN PAPER.

Variant	MuSiQue F1	2Wiki F1
ColBERTv2 (reference)	58.309	77.763
Learned HKVM calibration (reference)	56.754	88.391
Controller dense-only	58.247	77.679
Controller HKVM-only	56.760	88.415
Controller interaction-only	60.295	88.769
Controller rank-only	64.474	88.637
Controller score-only	65.254	88.696
Controller no-interaction	65.738	88.581
Controller full	65.073	88.846

TABLE VII

WHG-KV HYPERPARAMETER SENSITIVITY ON MUSIQUE (SEED 13). DEFAULT ROW USES THE PAPER-FACING CONFIGURATION. ALL VARIANTS REUSE THE FROZEN DEEPSEEK EVIDENCE TUPLE CACHE; F1 VALUES ARE SINGLE-SEED POINT ESTIMATES.

Group	Variant	AR@10	F1	EM
Default				
	$(\lambda_f{=}0.20, \lambda_s{=}0.40, \lambda_b{=}0.40)$, $\beta{=}2.0$, $\rho{=}0.35$	37.815	38.946	38.229
β *(weight scale)*				
	$\beta = 1.5$	37.360	38.693	37.940
	$\beta = 2.5$	38.395	39.482	38.767
	$\beta = 3.0$	38.395	39.834	39.139
	$\beta = 4.0$	39.015	39.918	39.222
ρ *(diffusion)*				
	$\rho = 0.15$	38.477	39.383	38.726
	$\rho = 0.25$	37.857	39.292	38.602
	$\rho = 0.45$	37.650	38.937	38.146
	$\rho = 0.55$	36.947	38.471	37.650
λ *(confidence weights)*				
	Uniform $(0.33, 0.33, 0.33)$	37.940	38.988	38.271
	Factual-only $(1, 0, 0)$	38.188	39.200	38.477
	Salience-only $(0, 1, 0)$	37.898	38.936	38.229
	Bridge-only $(0, 0, 1)$	37.940	39.022	38.271
	Factual+Bridge $(0.5, 0, 0.5)$	38.022	38.943	38.229

Across all 14 variants, WHG-KV F1 ranges from 38.471 to 39.918 (range 1.447), and all variants exceed the pairwise KG-PPR baseline (36.332 F1, Table 1 in the main paper). The default configuration is conservative: higher β and lower ρ modestly improve F1, but the paper's qualitative conclusions are robust to all tested hyperparameter perturbations. The default was chosen to avoid dev-set over-tuning rather than to maximize single-seed MuSiQue F1.

VII. RUN AND ARTIFACT PROVENANCE

The public artifact is split across two hosting platforms:

- **GitHub** (code, configs, scripts, results, manifests): https://github.com/Mingyu-Zh/HKVM-RAG
- **Hugging Face Dataset** (data, frozen outputs): https://huggingface.co/datasets/MingY-Zh/HKVM-RAG

This supplement itself is hosted in the GitHub repository at `supplemental_material.pdf`. Reviewers should download `data/` and `frozen_outputs/` from Hugging Face and place them at the repository root before running the paper-evidence verification script.

Table VIII lists the paper-facing result families and the artifact records required to verify them. The artifact includes scripts (`scripts/verify_paper_evidence.py`, `scripts/reproduce_fixed_substrate.py`), configs (`configs/`), and a SHA-256 file manifest (`manifests/FILES.sha256`) for integrity verification.

TABLE VIII

Paper-facing result provenance. The table identifies the protocol and artifact record for each result block; exact export roots, hashes, and redistributability constraints belong in the submitted artifact README.

Result block	Verifies	Prediction protocol	Bootstrap/export	Artifact record
P3 fixed-substrate matrix	Standalone WHG-vs-KG comparison and HotpotQA boundary	Frozen dev predictions; shared tuple cache; seeds 13/29/43	Seed-13 1,000-sample WHG-minus-KG bootstrap; paper table export	`results/paper_evidence/main_fixed_` `frozen_outputs/predictions/fixed_s`
Evidence-selection trajectory	Oracle headroom, support scorer, calibration, and controller sequence	Frozen dev predictions plus train-to-dev models; no dev labels at inference	Phase-specific paper exports; controller rows use 5,000-sample bootstrap	`results/paper_evidence/calibration` `results/run_summaries/`
Phase1D controller	Dense-aware controller vs ColBERTv2 and learned HKVM	OOF train-side HKVM predictions; frozen dev predictions; seeds 13/29/43	5,000-sample paired bootstrap over frozen prediction files	`results/paper_evidence/adaptive_so` `frozen_outputs/predictions/adaptiv`
Source-level diagnostics	WHG-KV source specificity under ColBERTv2/BM25/Contriever	First-stage source fixed per block; structured source swapped; no new LLM calls	5,000-sample baseline-vs-WHG bootstrap; non-WHG margins are point estimates	`results/paper_evidence/source_leve` bootstrap CSVs
Negative diagnostics	Naive-fusion and extraction-substrate boundaries	Diagnostic exports outside the frozen main matrix	Descriptive unless explicitly reported as paired bootstrap	`results/run_summaries/extraction_q` prompt/schema records